

Question Paper

Exam Date & Time: 30-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.Tech. (COMPUTER SCIENCE & ENGINEERING) EXAMINATIONS - DECEMBER 2024 / JANUARY 2025

SUBJECT: CSE 3121/CSE_3121 - PRINCIPLES OF CRYPTOGRAPHY
(MAKEUP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Analyze the AES encryption algorithm and propose a modern cryptographic technique that could enhance its security, considering the advancements in computing power and cryptanalysis techniques. Justify your choice and explain how it addresses the limitations of AES. (5)
- 1B) Compare and contrast CBC-based hash functions with modern hash functions like SHA-256 and SHA-512 in terms of security and efficiency. (3)
- 1C) Solve the following system of congruences: (2)
 $x \equiv 2 \pmod{3}$
 $x \equiv 3 \pmod{5}$
 $x \equiv 2 \pmod{7}$
using Chinese remainder Theorem and find the value of x .
- 2A) Determine the Public key, Private Key, using RSA algorithm for the given values $p=5$, $q=11$ and $e=3$. Verify the encryption and decryption process for $M=10$ using generated public and private key. (5)
- 2B) Using Linear Congruential Generator, generate random numbers for the values seed = 7, $a = 5$, $c = 3$ and $m = 11$. Evaluate the randomness of the sequence by analyzing any patterns or repeating cycles that might emerge in the generated numbers. (3)
- 2C) Identify how diffusion and confusion functions contribute to the security of the DES block cipher. (2)
- 3A) If the 4 key words of the 5th Round of AES-128 key expansion are 0x589D36EB, 0xFDEE387D, 0x0FCC9BED, and 0x4C4046BD respectively, calculate the 4 key words of the 6th Round. (4)
- 3B) Alice is generating a pair of prime numbers p and q for her RSA public-key cryptosystem using the following procedure: (4)
1) She randomly selects an odd integer starting from 1000 and finds that 1997 is prime. Analyze the implications of this choice for p in the context of RSA key generation.
2) Alice then randomly selects another odd integer, 1009, and the Miller-Rabin test confirms that it is prime. Compare the two prime numbers p and q , discussing their suitability for cryptographic applications.
3) Finally, compute the modulus n using the values of p and q that Alice has found. Evaluate how the size of n contributes to the overall security of the RSA algorithm.
- 3C) A Linear Congruential Generator (LCG) uses a modulus = 256, seed = 65, multiplier = 15, increment = 8, and has a period of 16. Calculate the 120th number of Random number sequence by excluding the seed (i.e. the first number of the sequence is X_1 which represents the usual notation). (2)
- 4A) Write any 4 Entropy sources for True Random Number Generators (TRNGs). Also, explain the seed requirements of Pseudorandom Number Generators (PRNGs) with relevant block diagram. (4)
- 4B) Compute the Decrypted Ciphertext (CT) = 'ZPGR' using Hill Cipher (By using only alphabets by ignoring the case as the character set) if the Key used for encryption was as shown in Figure 1. (3)

Analyze the decryption and explain any 1 corresponding disadvantage of using Hill Cipher.

$$\begin{bmatrix} 7 & 15 \\ 14 & 25 \end{bmatrix}$$

Figure 1:- Key (K) used for encryption using Hill Cipher

- 4C) Analyze the working of Hash Functions Based on Cipher Block Chaining. Elaborate with the use of equations, how hashing is achieved without the secret key? Also comment on the security aspects of Hashing based on CBC. (3)
- 5A) Write the SHA-512 round function by explaining its key components and operations. Discuss how these components interact to achieve the desired hashing properties. Deduce your observations made about the round function. (4)
- 5B) Compare six key storage encryption requirements for XTS-AES as outlined in the P1619 standard for securing data at rest. Assess how each requirement influences the overall security of stored data and determine their interdependencies (3)
- 5C) Explain the three primary applications for Public-Key Cryptosystems. For each category, describe the role of the keys involved in the cryptographic function and provide an example application for each category. (3)

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